

Lecture Notes on Differential Equations and Transform Techniques

Spring 2025

The purpose of these lecture notes is to provide a concise summary of the key concepts covered in the course. They are designed to serve as a **quick reference** and should be used in conjunction with the core textbooks listed below. While some sections are adapted from these referenced texts, the notes are not intended to replace the original works, which offer more comprehensive explanations. To gain a deeper understanding, students are encouraged to refer to the textbooks listed.

References:

1. W. E. Boyce, R. C. DiPrima, D. B. Meade, *Elementary Differential Equations and Boundary Value Problems*, 9th edition, Wiley, 1997.
2. Brad G. Osgood, *Lectures on Fourier Transform and Its Applications*, IAS/Park City Mathematics Series / American Mathematical Society (AMS), 2019.
3. Charles Henry Edwards, David E. Penney, *Differential Equations and Boundary Value Problems: Computing and Modeling*, 4th edition, Pearson, 2000.
4. George F. Simmons, *Differential Equations with Applications and Historical Notes*, 2nd edition, CRC Press, 2016.
5. Ruel V. Churchill, *Fourier Series and Boundary Value Problems*, McGraw-Hill, 1941.
6. Gustav Doetsch, *Introduction to the Theory and Application of the Laplace Transformation*, Springer, 2012.
7. Morris Tenenbaum, Harry Pollard, *Ordinary Differential Equations*, Dover Books on Mathematics, Dover Publications, 1985.

EE1060: Differential Equations and Transform Techniques**Lecture Notes - 14****Topics covered :**

1. General solution of Second-order Homogeneous ODE
2. Solution of Second-order Homogeneous ODE with constant coefficients

- Consider a homogeneous second-order ODE

$$\frac{d^2 y}{dx^2} + p(x) \frac{dy}{dx} + q(x)y = 0 \quad (1)$$

Let $y_1(x)$ denote one the solutions of the the ODE. denote a known solution of this equation. In order to find a second linearly independent solution, which is necessary for constructing the general solution, one possible method [1] is as follows: We assume that the second solution, $y_2(x)$ has the form $y_2(x) = v(x)y_1(x)$, where $v(x)$ needs to be determined. The goal is to determine $v(x)$ such that $y_2(x)$ also satisfies the original differential equation. It is important to note that, since $v(x)$ assumed to be a non-constant function, the solutions $y_1(x)$ and $y_2(x)$ will be linearly independent.

For the chosen $y_2(x)$, the first and second derivatives are

$$y_2'(x) = v'(x)y_1(x) + v(x)y_1'(x) \text{ and } y_2''(x) = v(x)y_1''(x) + 2v'(x)y_1'(x) + v''(x)y_1(x) \quad (2)$$

Since $y_2(x)$ is assumed to be solution, it must satisfy (1). Accordingly, using (2),

$$v(x)y_1''(x) + 2v'(x)y_1'(x) + v''(x)y_1(x) + p(x)[v'(x)y_1(x) + v(x)y_1'(x)] + q(x)[v(x)y_1(x)] = 0$$

$$\begin{aligned} & \xrightarrow{\quad \quad \quad 0} \\ v(x)[y_1''(x) + p(x)y_1'(x) + q(x)y_1(x)] + v''(x)y_1(x) + 2v'(x)y_1'(x) + p(x)v'(x)y_1(x) &= 0 \\ v''(x) + 2v'(x)\frac{y_1'(x)}{y_1(x)} + p(x)v'(x) &= 0 \quad (3) \\ \frac{v''(x)}{v'(x)} &= -2\frac{y_1'(x)}{y_1(x)} - p(x) \end{aligned}$$

Integrating on both sides gives

$$\ln|v'(x)| = -2\ln|y_1(x)| - \int p(x)dx \implies v'(x) = \frac{1}{y_1^2(x)} e^{-\int p(x)dx} \quad (4)$$

Integrating again on both sides

$$v(x) = \int \frac{1}{y_1^2(x)} e^{-\int p(x)dx} \quad (5)$$

It is important to keep in mind that only if $v(x)$ obtained from (5) is not a constant, the solutions $y_1(x)$ and $y_2(x)$ are linearly independent.

- **Solution of second-order homogeneous differential equations with constant coefficients:** Consider a second-order ODE with constant coefficients

$$\frac{d^2 y}{dx^2} + 2\zeta\omega_n \frac{dy}{dx} + \omega_n^2 y = 0^1 \quad (6)$$

One potential choice for solutions is the exponential function $y(x) = e^{sx}$, with a suitable choice of s . For e^{sx} to be the solution,

$$\frac{d^2 e^{sx}}{dx^2} + 2\zeta\omega_n \frac{de^{sx}}{dx} + \omega_n^2 e^{sx} = 0 \implies (s^2 + 2\zeta\omega_n s + \omega_n^2) e^{sx} = 0 \quad (7)$$

Since $e^{sx} \neq 0$, e^{sx} is a solution of 6 if and only if s satisfies the characteristic/auxiliary equation

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0 \quad (8)$$

Thus there are atmost two choices of s for which e^{sx} is a solution to (6).

$$s_1, s_2 = -\zeta\omega_n \pm \omega_n \sqrt{\zeta^2 - 1} \quad (9)$$

1. **Real and Distinct roots:** If $|\zeta| > 1$, the roots of the auxiliary equations are distinct. Thus the two functions $e^{s_1 x}$ and $e^{s_2 x}$ satisfy the 6. Furthermore, the two functions are also linearly independent since

$$W(e^{s_1 x}, e^{s_2 x}) = e^{s_1 x} (e^{s_2 x})' - (e^{s_1 x})' e^{s_2 x} = (s_2 - s_1) e^{(s_1 + s_2)x} \neq 0 \quad (10)$$

Thus the general solution of (6) for distinct roots is given by

$$y(x) = c_1 e^{s_1 x} + c_2 e^{s_2 x} \quad (11)$$

2. **Complex roots:** If $|\zeta| < 1$, the roots of the auxiliary equations are complex valued. A common way to represent the roots for this scenario is

$$s_1, s_2 = -\underbrace{\zeta\omega_n}_{\alpha} \mp \underbrace{\omega_n \sqrt{1 - \zeta^2}}_{\omega_d} \quad (12)$$

A potential candidate for the the general solution is

$$y(x) = c_1 e^{s_1 x} + c_2 e^{s_2 x} \quad (13)$$

However, since $s_2 = s_1^*$, it is necessary that $c_1 = c_2^*$ (recollect arguments covered in the course on Circuits and Network Analysis). Let $c = a + jb$ be the complex constant. Accordingly, (13) can be rewritten as

$$y(x) = (a + jb)e^{-\alpha x} e^{-j\omega_d x} + (a - jb)e^{-\alpha x} e^{j\omega_d x} = e^{-\alpha x} (2a \cos(\omega_d x) + 2b \sin(\omega_d x)) = R \cos(\omega_d x + \phi) \quad (14)$$

¹Standard form for this course

where $R = 2\sqrt{a^2 + b^2}$ and $\phi = \arctan\left(\frac{b}{a}\right)$. Thus the general solution of (6) for complex roots is given by

$$y(x) = R \cos(\omega_d x + \phi) \quad (15)$$

3. **Repeated roots:** When $\zeta = 1$, the roots of the characteristic equation repeat and are given by $s_1 = s_2 = -\omega_n$.

Under this scenario, one fundamental solution to ODE (6) is

$$y_1(x) = e^{-\omega_n x} \quad (16)$$

The other linearly independent solution can be computed using (??) as

$$y_2(x) = y_1(x)v(x); \text{ where } v(x) = \int \frac{1}{e^{-2\omega_n x}} e^{-\int -\omega_n dx} dx = \int dx = x \quad (17)$$

Thus the general solution of (6) for repeated roots is given by

$$y(x) = c_1 e^{sx} + c_2 x e^{sx} \text{ where } s = -\omega_n \quad (18)$$

References

- [1] G. F. Simmons, *Differential Equations with Applications and Historical Notes*, 2nd ed. Boca Raton, FL: CRC Press, 2016.